Limits to Growth in Uncerta Environments

Barnabé Ledoux,
David Lacoste
Laboratoire Gulliver - EPSCI
barnabe.ledoux@polytechnique.edu

Silvia Montagnani
iRisk - University of Lille
C3S - University of Frankfurt
silvia.montagnani.etu@univ-lille.fr

Background

Fluctuating environments impose a fundamental **trade-off between growth and survival**, affecting both ecological Neoclassical economic models often assume unlimited exponential growth, based on the premise of environmental stability an However, real-world growth processes unfold under conditions of inherent stochasticity and uncertainty. In biological systems strategies, including **bet-hedging through phenotypic diversification, dormancy** (such as desert plants' seeds and fungal spores and **adaptive sensing**, to buffer against environmental unpredictability. Financial systems and ecosystems alike can experience **minor perturbation triggers a catastrophic shift to a less favorable state.** For example, the Wall Street crash of 1929 or the 2008 shifts. Mechanisms that dampen small fluctuations (for instance, aggressive risk management that creates an illusion of stability of **a full-scale collapse** [1].

Motivations

extending **Kelly's gambling model** [2], we draw on the analogy between **ological survival strategies** and **investment strategies** in uncertain environments to:

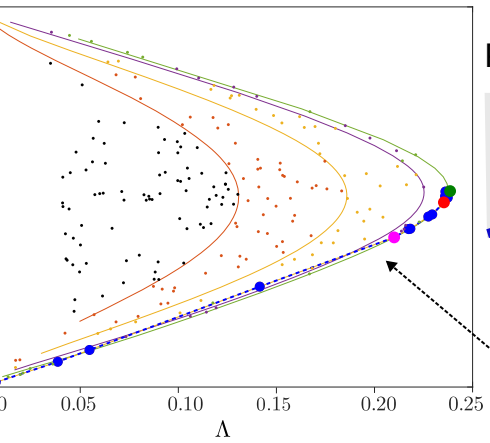
Explore **insights from biological adaptation strategies** for sustainable socioeconomic transitions.

Model debt-fueled growth as a stochastic process and the growth/risk trade-off.

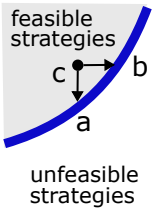
Quantify the **risk of collapse** due to debt.

Cre

amental trade-off between growth and risk in uncertain onments [3]: strategies on the Pareto front achieve the best promise between high average growth and low fluctuations.



Pareto optimality



A strategy in the Pareto front cannot be improved by moving in only one criterion in the optimization space

Pareto front

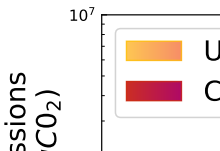
Growth/Risk Trade-Off

Here, we consider **a simple model of stochastic phenotypi switching between two phenotypes**, where organisms (such bacteria) grow in two randomly alterna ing environments. E phenotype is better adapted to one of these environment Analysing the trade-off between the average growth rate an variance, we uncover a "no risk, no gain" principle encapsula in a **Pareto front.** The shape of this front suggests that **a slig reduction in the maximum growth rate can significantly low risky fluctuations.** Moreover, the interplay between growth r and fluctuations is crucial in determining **extinction probabil** which **may increase with higher growth rates under certai conditions.**

Debt-Dependent Growth as a Financial Risk

Allowing leveraged bets dramatically increases potential growth and the risk of ruin. In simulations of the extended Kelly's model, **an agent who takes an initial loan to boost**

Capital Evolution with Borrowing

- C_τ be the capital at time τ,
- B_0 be the initial loan,

Predicted emissio consumption habits leverages.